

Project Proposal Group 30

COMPSCI/SFWRENG 4AL3: Applications of Machine Learning

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1 Overview

This project focuses on segmenting adult gliomas from MRI scans using a 2D U-Net convolutional neural network. Manual tumor segmentation requires trained raters to delineate tumor regions throughout a 3D MRI volume and is time-consuming. An automated model can generate an initial segmentation that can be reviewed and corrected by an expert, reducing the amount of manual delineation required. Automated segmentation is also useful for quantitative research because a fixed model applies the same learned segmentation procedure to every scan, providing more repeatable measurements across large datasets than repeated manual segmentation. This can make it easier to compare tumor regions across subjects or across scans collected at different time points.

Brain tumor segmentation is challenging because gliomas vary substantially between subjects in size, shape, location, and appearance. Tumor tissue can also resemble surrounding healthy tissue, and tumor regions may occupy only a small fraction of an MRI slice, creating a strong imbalance between tumor and background. In addition, different tumor sub-regions are distinguished by patterns across T1, T1CE, T2, and T2-FLAIR rather than by a single MRI sequence alone. The model must therefore combine information from the four aligned inputs while identifying irregular tumor boundaries and separating tissue classes with similar appearances.

2 Task Definition

2.1 Machine Learning Task

The project is a supervised semantic segmentation task. Rather than assigning one class to an entire MRI slice, the model performs dense multiclass classification by assigning a class to each spatial location in the slice. The ground-truth labels are provided by the expert-annotated segmentation masks included with the BraTS 2021 dataset. Each location is assigned exactly one class, making this a single-label, four-class segmentation problem rather than a multi-label classification problem.

2.2 Model Input

Each model input consists of four corresponding axial MRI slices from the same subject and anatomical position. The slices are taken from the T1, T1CE, T2, and T2-FLAIR volumes, which have already been co-registered so that the same spatial location corresponds to the same anatomy in each sequence. The four 240×240 slices are stacked as separate input channels, producing a $240 \times 240 \times 4$ input to the 2D U-Net.

Using all four sequences allows the model to use differences in tissue appearance across the MRI sequences when distinguishing tumor sub-regions. For example, T1CE provides information about contrast-enhancing tissue, while T2 and T2-FLAIR provide different contrast for tissue with increased water content, including regions associated with edema.

2.3 Model Output and Segmentation Classes

For each input slice, the model produces a 240×240 segmentation map with one predicted class for each spatial location. The original BraTS 2021 segmentation masks use labels 0, 1, 2, and 4. Before training, label 4 will be remapped to 3 so that the model predicts four consecutive classes:

- Class 0: background / non-tumor tissue
- Class 1: necrotic and non-enhancing tumor core (NCR/NET)
- Class 2: peritumoral edema / invaded tissue (ED)
- Class 3: enhancing tumor (ET), corresponding to label 4 in the original BraTS masks

The predicted 2D segmentation therefore corresponds directly to the labels from the matching axial slice of the original 3D BraTS segmentation mask.

2.4 Technical Challenges

The segmentation task is difficult because the four classes are highly imbalanced: background tissue occupies most of an MRI slice, while individual tumor sub-regions may contain relatively few pixels. Gliomas also vary substantially between subjects in their size, shape, location, and appearance, so the model must generalize across tumors with very different morphologies. Boundaries between tumor sub-regions may be irregular or visually ambiguous, requiring the model to distinguish tissue classes using information distributed across the four MRI sequences. Finally, because the proposed model operates on individual 2D slices, it does not directly use information from adjacent slices in the 3D MRI volume.

3 Data Source(s) and Plan for Data Collection

3.1 Primary Data Source, Access, and Licensing

The primary data source is the **BraTS 2023 Adult Glioma (GLI) dataset**, distributed through Synapse (Project ID syn51156910) and originally described by the RSNA-ASNR-MICCAI BraTS Challenge [1, 7, 4]. This dataset reuses the same adult glioma cohort as the earlier BraTS 2020/2021 challenges, but relabels the enhancing tumor class from 4 to 3, so no post-hoc label remapping is required on our end. Each subject provides four co-registered MRI volumes (T1, T1CE, T2, T2-FLAIR) and an expert-annotated 3D ground-truth mask. We use 2D axial slices from these volumes as input to a 2D U-Net, which predicts a 2D tumor sub-region segmentation mask per slice.

3.1.1 Access points

- **Synapse (primary, challenge portal):** <https://www.synapse.org/#!/Synapse:syn51156910>. Requires registration and acceptance of the BraTS data usage agreement.
- **TCIA (archival mirror of the same cohort):** DOI <https://doi.org/10.7937/jc8x-9874>, hosting the processed NIfTI volumes under a CC-BY 4.0 license. Related TCGA glioma collections are described in [2, 3].

- **Kaggle** (convenient for download, but secondary): dataset page [5]. Related uploads such as nagisa0v0/brats2023-gli-training-data are available.

3.1.2 Terms of Use

No web scraping is performed. Data is obtained through the authenticated Synapse clients or the Kaggle mirror, all of which are static bulk downloads with no API rate limiting to manage. The data is licensed for non-commercial research use with attribution (CC-BY 4.0 via TCIA). All data was de-identified by the organizers prior to release, and we will not attempt to re-identify subjects. We will include the required acknowledgement referencing the BraTS organizers and Synapse project, with citations provided in the shared bibliography.

3.2 Collection, Annotation and Pre-processing

3.2.1 Acquisition:

Retrospective, routine clinical pre-operative multiparametric MRI (mpMRI) of adult gliomas (WHO grades II–IV), contributed by multiple institutions with varying scanners, field strengths (1.5T/3T), and protocols. This heterogeneity is intentional and tests model generalization across acquisition conditions.

Modalities: native T1, post-contrast T1-weighted (T1CE/T1Gd), T2-weighted, and T2-FLAIR per subject.

3.2.2 Annotation protocol:

Labels were produced by manual delineation by 1 - 4 trained raters following the standardized BraTS protocol, then reviewed and approved by experienced neuroradiologists. **We do not annotate any data ourselves.** Manual voxel-level delineation of a 3D glioma volume is a labor-intensive expert task, which is why we use this pre-annotated benchmark rather than labeling by hand. The released labels are:

Label	Region	Abbrev.
0	Background / non-tumor	—
1	Necrotic and non-enhancing tumor core	NCR/NET
2	Peritumoral edema / invaded tissue	ED
3	GD-enhancing tumor	ET

(Note: In BraTS 2023 dataset the enhancing-tumor label is 3 rather than the historical value 4, so no remapping is required.) Standard evaluation regions remain Whole Tumor = $\{1, 2, 3\}$, Tumor Core = $\{1, 3\}$, Enhancing Tumor = $\{3\}$.

3.2.3 Pre-processing

Preprocessing applied by the organizers: DICOM to NIfTI conversion (with PHI removed), rigid co-registration of the four modalities, registration to the SRI24 anatomical template, resampling to 1 mm³ isotropic resolution, and skull-stripping. Volumes are $240 \times 240 \times 155$ voxels stored as `.nii.gz`.

Preprocessing performed by us: Intensities have no universal scale in MRI, so we will apply per-volume, per-modality z-score normalization computed over non-zero (brain) voxels only, optionally with percentile clipping. We will then stack the four modalities as channels, extract axial 2D slices, and apply training-time augmentation (flips, small rotations, elastic deformation, intensity shifts) identically across channels and mask.

3.3 Available Metadata

Each case is identified by a subject ID (e.g., BraTS-GLI-00000-000). A subset of cases includes a crosswalk to the original TCIA/TCGA identifiers and MGMT promoter methylation status (used in the separate radiogenomic task). Metadata is not used as a model input. In this project, the model input is the four imaging modalities only.

3.4 Dataset Size, Splits and Use

- **3D volumes:** 1251 publicly labeled training subjects. Each contributes four modality volumes and one mask. A further 219 validation cases are released with hidden labels and cannot be used for local evaluation, so all train/validation/test splits are drawn from the 1251 labeled subjects.
- **2D slices:** Each subject has 155 axial slices, giving roughly 194,000 slices in total. After discarding empty (all-zero) slices and subsampling slices without tumor, we expect on the order of tens of thousands of multi-channel training slices. The exact count will be computed after download. This far exceeds the 1000-data-point minimum.

- **Entire dataset vs subset:** We plan to use the entire labeled cohort. Tumor voxels form a small minority of each volume, so every tumor-containing slice is valuable. Inter-subject and inter-scanner variability also improves generalization. If memory or compute limits arise during prototyping, a random subject-level subset (30–50%) may be used temporarily, with final models trained on the full set.
- **Splitting:** The train/validation/test split ($\approx 70/15/15$) is performed strictly by subject to avoid leakage between adjacent slices of the same patient.

3.5 Example Data Points

Each data point for the 2-D U-Net is a 4-channel 240×240 (or cropped/resized) axial slice together with its corresponding integer label map.

1. **Subject BraTS-GLI-00003-000, axial slice $z = 103$** Input: stacked [T1, T1CE, T2, FLAIR]. Label map: a solid necrotic/non-enhancing core (Label1) at the center, surrounded by a thin ring of enhancing tumor (Label3), with a much larger region of peritumoral edema (Label2) extending outward into the surrounding brain tissue.
2. **Subject BraTS-GLI-00019-000, axial slice $z = 65$** Input: stacked [T1, T1CE, T2, FLAIR]. Label map: a small necrotic core (Label1) sits inside a larger enhancing tumor region (Label3), which itself is embedded within a much larger zone of peritumoral edema (Label2) that spreads further into the surrounding brain tissue. The core is proportionally smaller relative to the enhancing and edema regions.
3. **BraTS-GLI-00838-000, axial slice $z = 81$** Input: stacked [T1, T1CE, T2, FLAIR]. Label map: Two separate tumor nodules present on the same slice, each with its own small necrotic core (Label1) wrapped by a thin enhancing rim (Label3), both sitting within one large shared region of edema (Label2) representing multifocal/multiple tumor foci within one edema region.

4 Solution

U-Net is a CNN based architecture designed for image segmentation tasks. This architecture consists of an encoder and decoder. Skip connections carry features from the encoder to the decoder to help recover spatial detail. [9]

Following the TA’s advice, we will compare a simpler CNN-based segmentation model with a 2D U-Net to investigate whether the more complex architecture improves tumor segmentation enough to justify its computational cost. It should be noted that both models are CNNs; however, the baseline will use a small number of convolutional and pooling layers followed by upsampling and a pixel-wise classification layer, but without encoder–decoder skip connections (as seen in U-Net.) This baseline is inspired by fully convolutional networks, which produce dense segmentation maps rather than a single label for an image [6].

Both models will use the same four-channel axial slices and four-class segmentation masks described above. The split will be done by patient/subject, not by individual slice. Otherwise adjacent slices from the same patient could appear in both training and testing which will artificially inflate results. Both models will use the same approximately 70/15/15 train/validation/test split, preprocessing, augmentation, and comparable training and tuning budgets. Model settings will be selected using validation performance, with the test set reserved for the final evaluation.

Dice score will be used as the main segmentation metric. This is because in the context of tumor segmentation, background pixels will vastly outnumber tumor pixels. Unlike conventional pixel wise accuracy, Dice measures the overlap between predicted and ground truth masks without rewarding correct background predictions. As outlined in the data section, it will be evaluated for whole tumor, tumor core, and enhancing tumor.

We will compare cross-entropy loss with a combined Dice + cross-entropy loss for both architectures, giving a total of four model/loss combinations. The Dice-based objective is inspired by V-Net’s approach to foreground/background imbalance in a similar medical image segmentation task [8].

Although our models will operate on 2D slices, since the MRI has 155 axial slices, we will calculate Dice across the entire reconstructed 3D volume (as opposed to calculating Dice on each 2D slice separately and averaging the scores.) This is because some slices will typically contain no tumor at all, while others will contain large tumor regions. We will also inspect predicted masks for missed tumors and inaccurate boundaries, and record parameter counts and inference time on the same hardware. These controlled comparisons will assess architecture and loss choices on our own splits rather than simply reproduce a published model.

Relevant libraries are PyTorch, NumPy, NiBabel (for reading nii.gz data files).

5 TA Consultation

We consulted Yasmin Ahmadi regarding our proposed MRI segmentation project using the BraTS 2021 dataset. We discussed the available data, including the four MRI modalities (T1, T1CE, T2, and T2-FLAIR), and our plan to extract 2D axial slices from the 3D volumes for model training. We also discussed the target segmentation labels and our plan to use a U-Net architecture.

Yasmin recommended establishing a simpler CNN-based segmentation model as a baseline against which to evaluate U-Net. She suggested comparing models of increasing complexity to determine whether a more advanced model provides a meaningful improvement in segmentation performance, rather than evaluating U-Net in isolation. She also suggested considering binary segmentation depending on the final definition of our target labels.

Based on this feedback, we plan to retain the four-class segmentation task defined by the BraTS annotations while introducing a simpler CNN-based segmentation model as our baseline. We will then implement U-Net and evaluate both models using the same patient-level training, validation, and testing splits. We plan to use the Dice similarity coefficient to measure overlap between predicted and ground-truth tumor regions. This will allow us to directly compare U-Net against the simpler CNN baseline and determine whether the additional architectural complexity produces a meaningful improvement in segmentation performance.

References

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